15 Lecture 15 (February 19, 2025, est.): Schur functors and Deligne's theorem

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Throughout this lecture, k is algebraically closed of characteristic zero. We follow the framework of [Del02]. For part of the discussion, one may drop the finite-length and finite-dimensional-Hom conditions: retain an essentially small k-linear abelian rigid symmetric monoidal category $\mathcal{C}$ with $\mathrm{End}(\mathbb{1}) = k$. Tensor product is exact in each variable, because rigidity supplies both adjoints. These are still substantive hypotheses; retaining rigidity alone would not suffice.

15.1 The statement and the finiteness condition

15.1 Definition. An object $X \in \mathcal{C}$ is of *moderate growth* if its tensor powers have finite length and there is $a_X \geq 1$ with

$$\mathrm{length}(X^{\otimes n}) \leq a_X^n \qquad (n \geq 1).$$

It is *Schur-finite* if $S^\lambda(X) = 0$ for some partition λ, where the Schur functor is defined below.

15.2 Theorem (Deligne)**.** *An object is Schur-finite if and only if it is of moderate growth. Schur-finite objects are closed under finite direct sums, tensor products, duals, extensions and subquotients. A symmetric tensor category has a fiber functor to* SVec_k *if and only if all its objects are Schur-finite.*

The existence of the super fiber functor is the deep part of this theorem. This lecture establishes the Schur decomposition and the implication from moderate growth to Schur-finiteness, and records the representation-theoretic tools used in the full proof. The remaining arguments are in [Del02, Proposition 0.5 and Theorem 0.6]; they are not consequences of the growth estimate alone.

15.3 Lemma. *If every object of* $\mathcal{C}$ *has finite length, then every morphism space is finite-dimensional.*

Proof. We use the standard simple-unit lemma for rigid abelian tensor categories: every subobject of $\mathbb{1}$ is a direct summand, so $\mathrm{End}(\mathbb{1}) = k$ implies that $\mathbb{1}$ is simple [DM82, Proposition 1.17]. This is the input used in [Del02, Proposition 1.1]; we do not prove that lemma here. For an object Z of finite length, a linearly independent family of maps $\mathbb{1} \to Z$ gives a monomorphism from the corresponding finite direct sum of copies of $\mathbb{1}$ to Z: any kernel is a direct sum of copies of the simple object $\mathbb{1}$ and would give a linear relation. Thus $\dim_k \mathrm{Hom}(\mathbb{1}, Z) \leq \mathrm{length}(Z)$. Finally rigidity gives

$$\mathrm{Hom}(X, Y) \cong \mathrm{Hom}(\mathbb{1}, X^* \otimes Y),$$

and $X^* \otimes Y$ has finite length by assumption. ∎

15.2 Finite-group actions inside a category

For a finite-dimensional vector space V and $X \in \mathcal{C}$, the notation $V \otimes X$ denotes the corresponding finite direct sum of copies of X, characterized independently of a choice of basis. Define

$$\underline{\mathrm{Hom}}(V, X) = V^* \otimes X.$$

Suppose a finite group G acts on X, that is, we have a homomorphism $G \to \mathrm{Aut}_{\mathcal{C}}(X)$. The averaging operator

$$e_G = \frac{1}{|G|} \sum_{g \in G} g \in \mathrm{End}_{\mathcal{C}}(X)$$

is idempotent. Its image is the invariant object X^G. Its kernel is the sum of the images of $g - 1$, so the coinvariant object $X_G = \mathrm{coker}(\bigoplus_{g \in G} X \xrightarrow{(g-1)} X)$ is canonically isomorphic to $\operatorname{im} e_G$.

If $V \in \mathrm{Rep}_k(G)$, there is a diagonal action on $V^* \otimes X$, and we put

$$\underline{\mathrm{Hom}}_G(V, X) = (V^* \otimes X)^G.$$

This is an object of $\mathcal{C}$, rather than merely the vector space of morphisms from V to X.

15.4 Proposition (Isotypic decomposition)**.** *Let $(V_\lambda)_{\lambda \in I}$ be representatives of the irreducible kG-modules. For every object X carrying a G-action, evaluation induces a natural G-equivariant isomorphism*

$$\bigoplus_{\lambda \in I} V_\lambda \otimes \underline{\mathrm{Hom}}_G(V_\lambda, X) \xrightarrow{\sim} X.$$

Proof. Maschke's theorem and the algebraic closure of k give $kG \cong \prod_{\lambda \in I} \mathrm{End}_k(V_\lambda)$. The central idempotents split X into isotypic summands. In the λ-summand, matrix units split it into $\dim V_\lambda$ isomorphic summands; their multiplicity object is precisely $(V_\lambda^* \otimes X)^G$. The resulting isomorphism is evaluation, so it is independent of the matrix units chosen. ∎

15.3 Partitions and Schur functors

A partition of n is a finite sequence $\lambda = (\lambda_1, \ldots, \lambda_r)$ with $\lambda_1 \geq \cdots \geq \lambda_r > 0$ and $|\lambda| = \sum_i \lambda_i = n$. Its Young diagram is

$$[\lambda] = \{(i, j) : 1 \leq i \leq r,\ 1 \leq j \leq \lambda_i\}.$$

The transpose λ^{t} is defined by interchanging the two coordinates. For example, $(3, 1)^{\mathrm{t}} = (2, 1, 1)$. We also allow the empty partition of 0, with Schur functor equal to the constant unit object; this convention is used in the direct-sum formula below. The irreducible representations of S_n are indexed by partitions of n; write V_λ for the corresponding representation.

The symmetry on $\mathcal{C}$ defines an action of S_n on $X^{\otimes n}$ by adjacent transpositions. The symmetric, rather than merely braided, condition ensures that these transpositions square to the identity.

15.5 Definition. For $\lambda \vdash n$, the *Schur functor* is

$$S^\lambda(X) = \underline{\mathrm{Hom}}_{S_n}(V_\lambda, X^{\otimes n}).$$

Applying the isotypic decomposition gives

$$X^{\otimes n} \cong \bigoplus_{\lambda \vdash n} V_\lambda \otimes S^\lambda(X). \tag{15.1}$$

For the one-row and one-column partitions, $V_{(n)}$ is trivial and $V_{(1^n)}$ is the sign representation. Consequently

$$S^{(n)}(X) = \mathrm{Sym}^n(X), \qquad S^{(1^n)}(X) = \bigwedge^n X.$$

In characteristic zero these are the images of the symmetrizing and antisymmetrizing idempotents, respectively.

15.6 Proposition. *If X is of moderate growth, then some Schur functor annihilates X.*

Proof. Suppose instead that $S^\lambda(X) \neq 0$ for every partition λ. Each summand in (15.1) then contributes at least $\dim V_\lambda$ to the length. Therefore

$$\mathrm{length}(X^{\otimes n}) \geq \sum_{\lambda \vdash n} \dim V_\lambda \geq \left(\sum_{\lambda \vdash n} (\dim V_\lambda)^2 \right)^{1/2} = \sqrt{n!}.$$

The last equality is the dimension of the regular representation of S_n. For even n, $n! \geq (n/2)^{n/2}$, so $(\sqrt{n!})^{1/n} \geq (n/2)^{1/4}$ tends to infinity. This contradicts any exponential bound a_X^n. ∎

15.7 Example. For an ordinary vector space of dimension d, $\bigwedge^{d+1} V = 0$. An odd one-dimensional supervector space instead has $\mathrm{Sym}^2(V) = 0$, while its exterior powers never vanish. Thus exterior powers alone cannot characterize super fiber functors. Schur functors allow both kinds of vanishing.

15.4 Restriction, induction, and multiplicities

For $n = n_1 + \cdots + n_r$, the Young subgroup $S_{n_1} \times \cdots \times S_{n_r} \subseteq S_n$ acts on consecutive blocks of letters. Its irreducibles are the external tensor products $V_{\mu_1} \boxtimes \cdots \boxtimes V_{\mu_r}$, with $\mu_i \vdash n_i$. Define

$$[\lambda : \mu_1, \ldots, \mu_r] = \dim_k \mathrm{Hom}_{\prod_i S_{n_i}} \left(\boxtimes_i V_{\mu_i}, \mathrm{Res}^{S_n}_{\prod_i S_{n_i}} V_\lambda \right).$$

Frobenius reciprocity identifies this with the multiplicity of V_λ in $\mathrm{Ind}^{S_n}_{\prod_i S_{n_i}}(\boxtimes_i V_{\mu_i})$. These nonnegative integers are computed by the Littlewood–Richardson rule, iterated when $r > 2$.

For example, when $r = 2$ the same decomposition yields

$$S^\lambda(X \oplus Y) \cong \bigoplus_{|\mu|+|\nu|=|\lambda|} (S^\mu(X) \otimes S^\nu(Y))^{\oplus [\lambda:\mu,\nu]}.$$

Indeed, expand $(X \oplus Y)^{\otimes n}$ by the subsets of tensor positions occupied by X; each fixed cardinality is induced from its Young subgroup, and then apply the multiplicity-object functor. Likewise,

$$S^\mu(X) \otimes S^\nu(X) \cong \bigoplus_{\lambda \vdash |\mu|+|\nu|} S^\lambda(X)^{\oplus [\lambda:\mu,\nu]}.$$

These formulas explain how the representation theory of symmetric groups controls the operations appearing in the Schur-finiteness criterion.

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